



A USSD-Based Smart Agricultural Advisory and Harvest Prediction System to Enhance Food Security Among Smallholder Farmers in Rwanda

BSc. in Software Engineering

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LIST OF ACRONYMS AND ABBREVIATIONS

AI Artificial Intelligence

ALU African Leadership University

API Application Programming Interface

ERD Entity-Relationship Diagram

FAO Food and Agriculture Organization (of the United Nations)

GDP Gross Domestic Product

IoT Internet of Things

IPCC Intergovernmental Panel on Climate Change

JWT JSON Web Token

MAE Mean Absolute Error

MINAGRI Ministry of Agriculture and Animal Resources
(Rwanda) **ML** Machine Learning

MLOps Machine Learning Operations

NISR National Institute of Statistics of Rwanda

REST Representational State Transfer

RMSE Root Mean Square Error

RURA Rwanda Utilities Regulatory Authority

RWF Rwandan Franc

SDLC Software Development Life Cycle

SDG Sustainable Development Goal

SMS Short Message Service

SUS System Usability Scale

UML Unified Modeling Language

USSD Unstructured Supplementary Service Data

UUID Universally Unique Identifier

XGBoost Extreme Gradient Boosting

ABSTRACT

Smallholder farmers in Rwanda contribute the majority of national food production yet remain disproportionately exposed to climate-driven yield variability and lack access to timely, localised agronomic information. While smartphone-based advisory platforms exist, the persistent feature phone majority and limited rural internet penetration leave the most vulnerable farmers behind. This research proposes the design and development of a USSD-based smart agricultural advisory and harvest prediction system that couples a machine-learning yield model with a low-bandwidth menu interface accessible on any GSM handset. The study follows an iterative agile-scrum methodology over nine weeks (24 May - 28 July 2026): historical crop and weather datasets from NISR, MINAGRI and the Rwanda Meteorology Agency are used to train and benchmark Random Forest, XGBoost and Multilayer Perceptron regressors for beans and maize, while a FastAPI backend exposes the model through a USSD sandbox simulated via Africa's Talking. A retrospective validation against 2020-2024 yield records and a usability pilot with 10-15 farmers in Nyagatare District will measure technical and human-centred outcomes. The system is hypothesised to achieve at least 70% yield-prediction accuracy ($R^2 \geq 0.70$) and a System Usability Scale score of ≥ 75 among pilot participants, demonstrating that affordable, AI enabled advisory services can be delivered on the phones farmers already own.

Keywords: USSD, machine learning, harvest prediction, smallholder farmers, Rwanda, food security, feature phone, agricultural advisory.

CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

Agriculture remains the structural backbone of Rwanda's economy and the livelihood of more than 65% of the national workforce, contributing approximately 25-30% of gross domestic product (National Institute of Statistics of Rwanda [NISR], 2023). Across Sub-Saharan Africa, smallholder farmers cultivating fewer than two hectares produce close to 80% of the food supply, making their productivity directly determinant of regional food security (Food and Agriculture Organization [FAO], 2021). Within Rwanda, the Eastern Province, and Nyagatare District in particular, functions as a critical breadbasket for staple crops such as beans and maize, supplying both subsistence and commercial markets.

However, this productive base is increasingly destabilised by climate variability. Recent assessments by the Intergovernmental Panel on Climate Change (IPCC, 2022) identify East Africa as a hotspot for rainfall irregularity, with the short rainy season (Season A) shifting by up to three weeks compared with the historical mean. Downscaled projections released by the Rwanda Meteorology Agency (2022) confirm increasing intra-seasonal variability, longer dry spells and higher temperature extremes in Nyagatare. Such conditions undermine the traditional calendar-based planting decisions that most smallholders still rely on.

Traditional advisory channels in Rwanda combine extension officers, radio programmes and printed bulletins. While effective in scope, they cannot deliver the per-farm, per-week granularity now required, and they offer limited interactivity. Digital tools such as the e-Soko price platform, PlantVillage Nuru and various smartphone-based weather applications have narrowed this gap, but their reach is constrained by smartphone penetration which remained at 26.7% of mobile subscribers in 2023, compared with universal access to basic 2G handsets (Rwanda Utilities Regulatory Authority [RURA], 2023). Unstructured Supplementary Service Data (USSD), the protocol that underpins mobile money services such as MTN MoMo *182# and Airtel *500#, has demonstrated that complex digital transactions can be delivered to any GSM handset at near-zero marginal cost (Aker & Mbiti, 2010). This research argues that the same channel can carry machine-learning-driven agricultural advisories to the farmers who need them most.

1.2 Problem Statement

First, there is an information asymmetry between modern climate-smart agricultural knowledge and the channels available to rural smallholders. Studies in West and East Africa show that farmers without timely, localised weather and agronomic information experience yield losses ranging from 15% to 40% during anomalous seasons (Aker & Fafchamps, 2015; Lipper et al., 2014). In Rwanda, MINAGRI's own digital initiatives report (2021) recognises that the e Soko platform addresses market prices but does not provide predictive, plot-specific advisories.

Second, the existing software solutions that do incorporate machine learning, including PlantVillage Nuru and FAO's WaPOR dashboard, presuppose smartphone ownership, app-store access and a mobile data subscription (Hughes et al., 2021). Given that 70% of mobile devices in rural Rwanda are feature phones, these tools systematically exclude the very population most exposed to yield risk, producing what Aker and Mbiti (2010) describe as a 'digital divide within the digital divide'.

Third, even where USSD-based agricultural services have been piloted in the region, such as iCow in Kenya and Esoko in Ghana, they have remained largely rule-based, broadcasting generic seasonal advice rather than personalised, model-driven predictions. There is therefore a documented gap for a system that integrates supervised machine-learning yield prediction with a USSD interface tailored to local crops, languages and agro-ecological zones.

This research proposes a USSD-based smart agricultural advisory and harvest prediction system that integrates: (i) a Random Forest / Gradient Boosting yield prediction model trained on NISR and MINAGRI historical data fused with Rwanda Meteorology Agency forecasts; (ii) a USSD menu interface in English and Kinyarwanda; and (iii) a rules-and-ML hybrid advisory engine. The system targets the three problems above by delivering personalised, AI-derived insights through the channel and language already universal among Rwandan smallholders.

1.3 Project Main Objective

To design, develop and pilot a USSD-based smart agricultural advisory and harvest prediction

system that delivers localised weather insights, planting guidance and machine-learning-based yield estimates to smallholder farmers in Nyagatare District, Rwanda, in order to support informed decision-making and strengthen household food security.

1.3.1 Specific Objectives (SMART)

1. To review existing literature on machine-learning yield prediction and USSD-based agricultural advisory systems, and to acquire historical crop and weather datasets for Nyagatare District from NISR, MINAGRI and the Rwanda Meteorology Agency, completed by 21 June 2026.
2. To design and prototype a bilingual (English / Kinyarwanda) USSD menu interface and a FastAPI backend integrated with a live weather API, deployed to an Africa's Talking sandbox by 5 July 2026.
3. To train and benchmark at least three supervised regression models (Random Forest, XGBoost, Multilayer Perceptron) for bean and maize yield prediction, achieving R squared of at least 0.70 and Mean Absolute Error within 15% of the historical mean yield, by 12 July 2026.
4. To conduct a usability pilot with 10-15 farmers in Nyagatare District using simulated USSD sessions plus retrospective yield validation, targeting a System Usability Scale (SUS) score of at least 75 and task-completion rate of at least 80%, between 12 and 22 July 2026.
5. To analyse pilot results and produce a final report and defence presentation evidencing the system's contribution to food-security decision support, by 28 July 2026.

1.4 Research Questions

6. How can USSD technology be designed to deliver predictive, machine-learning-derived agricultural advisories to Rwandan smallholder farmers using feature phones?
7. Which supervised regression algorithm (Random Forest, XGBoost or Multilayer Perceptron) achieves the highest accuracy when predicting bean and maize yields in Nyagatare District using fused NISR, MINAGRI and meteorological data?
8. To what extent does the availability of localised weather and ML-based yield information change the stated planting and harvesting decisions of pilot farmers?

9. What usability, language and adoption challenges emerge when deploying a USSD-based predictive advisory service in rural Rwanda, and how can these be mitigated in the system's design?

1.5 Project Scope

Geographically, the study is confined to Nyagatare District in the Eastern Province of Rwanda, a high-priority agricultural zone with documented climate vulnerability. Crop-wise, the scope is limited to two staple crops, common beans (*Phaseolus vulgaris*) and maize (*Zea mays*), which together represent over 60% of Nyagatare's cultivated area (MINAGRI, 2021).

On the user side, the pilot targets 10-15 smallholder farmers recruited through a local cooperative. This sample size is smaller than the 50 farmers initially considered, in recognition of the nine-week project window and the impracticality of observing a full planting season; model accuracy is therefore validated against historical NISR yield records (2020-2024) in addition to the live pilot.

Technically, the USSD service is deployed in Africa's Talking sandbox rather than through a commercial telecom shortcode. The system excludes Internet-of-Things soil sensors, satellite-imagery ingestion, and crop-disease image diagnostics, which are noted as future work.

1.6 Significance and Justification

For farmers, the proposed system promises to reduce the information deficit that currently translates into preventable yield losses. By delivering personalised advisories on a channel that requires no smartphone, no data plan and no literacy in app-store navigation, the research directly addresses an equity gap in Rwanda's digital agriculture agenda.

For policymakers and the Ministry of Agriculture, the project demonstrates that machine learning capabilities can be retrofitted onto the universally available USSD channel, providing a low-cost blueprint for scaling AI-enabled extension services nationwide. Aligned with Sustainable Development Goal 2 (Zero Hunger) and Rwanda's Vision 2050 priority of climate smart agriculture, the system contributes a reproducible model for other Sub-Saharan African

contexts.

For the academic community, the project contributes empirical evidence on the integration of supervised ML regressors with low-bandwidth telecommunication protocols, an under-researched intersection in software engineering for development.

1.7 Research Budget

The project is deliberately scoped to rely on free-tier and open-source components wherever possible. Table 1.1 summarises the costs that must be funded, all of which are non-substitutable.

Item Description Qty Unit Cost Total

				(USD)	(USD)
1	Africa's Talking USSD sandbox credits	Simulated shortcode for end-to end testing	1 month	10.00	10.00
2	Cloud hosting (Render / Railway free + paid tier)	FastAPI backend + PostgreSQL during pilot	2 months	7.50	15.00
3	Rwanda Meteorology Agency API (paid quota)	District-level forecasts beyond free tier	1	20.00	20.00
4	Pilot participant airtime stipends	RWF 2,000 x 15 farmers	15	1.60	24.00
5	Field travel to Nyagatare (round	Bus fare + lodging, 2 visits	2	35.00	70.00

	trip)				
6	Translation review (Kinyarwanda)	Native speaker, USSD copy + SMS templates	1	30.00	30.00
7	Printing and stationery	Consent forms, usability scripts	1	10.00	10.00
8	Contingency (10%)	Unforeseen expenses	1	17.90	17.90

Table 1.1: Research Budget (USD). Estimated total: USD 196.90 (~RWF 246,000 at 1 USD = 1,250 RWF).

1.8 Research Timeline

The project executes in nine weeks between 24 May and 28 July 2026. Figure 7 presents the Gantt chart and Table 1.2 summarises the major milestones.

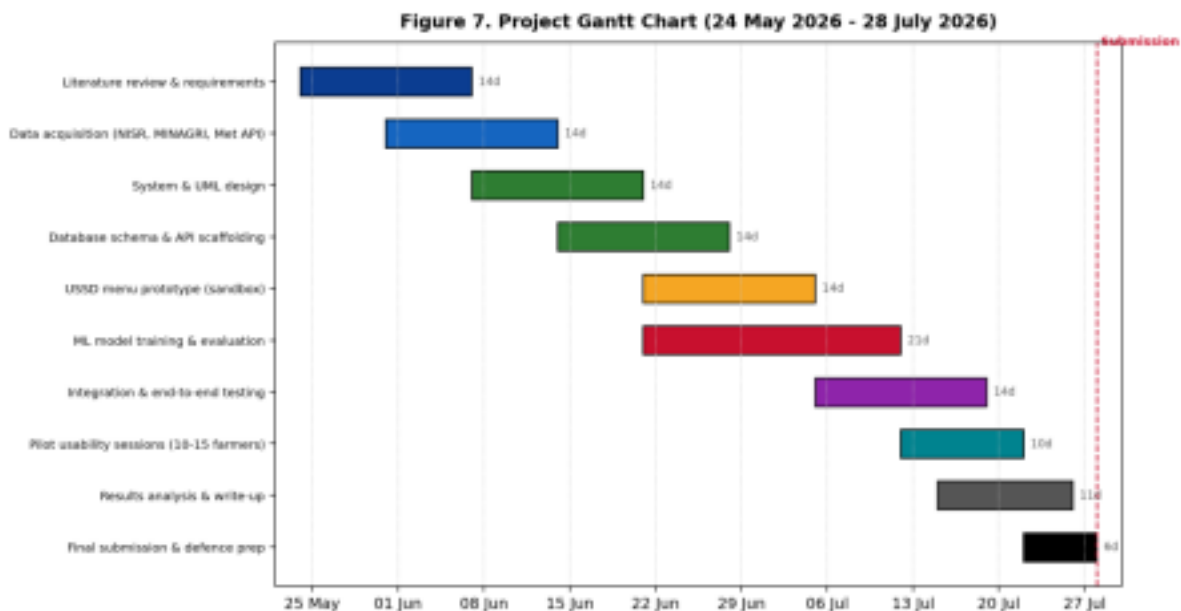


Figure 7: Project Gantt Chart (24 May - 28 July 2026).

Week	Dates	Key Deliverable
1-2	24 May - 6 Jun	Literature review complete; datasets acquired
3	7 Jun - 13 Jun	System design and UML diagrams finalised
4	14 Jun - 20 Jun	Database schema and FastAPI scaffolding
5	21 Jun - 27 Jun	USSD menu prototype on sandbox; ML training begins
6	28 Jun - 4 Jul	Best model selected; integration with USSD handler
7	5 Jul - 11 Jul	End-to-end testing in bilingual mode
8	12 Jul - 18 Jul	Field pilot with farmers in Nyagatare
9	19 Jul - 28 Jul	Results analysis, final report, defence prep

Table 1.2: Project Schedule Summary.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews the software-engineering and applied machine-learning literature relevant to mobile-based agricultural advisory systems and crop-yield prediction. The search targeted peer-reviewed journal articles, conference papers and credible institutional reports published between 2018 and 2025. Sources were drawn from IEEE Xplore, ACM Digital Library, ScienceDirect (Elsevier), MDPI, Springer, Google Scholar and the FAO and World Bank document repositories. The search combined the keyword clusters {USSD OR feature phone OR mobile advisory} AND {agriculture OR smallholder OR crop} AND {machine learning OR yield prediction OR random forest OR XGBoost}, restricted to English and to Sub Saharan or East African contexts where possible. After de-duplication and abstract screening, 22 papers met the inclusion criteria of software-related contribution and relevance to predictive advisory systems for resource-constrained users, of which 12 are discussed in depth in this chapter.

2.2 Historical Background of the Research Topic

The convergence of mobile telephony and agricultural advisory services dates to the mid 2000s, when SMS-based market-price platforms such as Esoko (Ghana, 2005), M-Farm (Kenya, 2010) and Rwanda's e-Soko (2012) demonstrated that text messaging could lower information asymmetry in rural markets (Aker & Mbiti, 2010). The USSD protocol, originally defined in 3GPP TS 22.090 for telecom signalling, was popularised in Africa by mobile-money services from 2007 onwards; its session-based, menu-driven, network-billed nature made it uniquely suited to low-resource users (RURA, 2023).

In parallel, the application of machine learning to agricultural yield prediction matured from regression-on-weather baselines in the early 2010s to ensemble and deep-learning models capable of integrating remote-sensing, soil and meteorological signals (van Klompenburg, Kassahun, & Catal, 2020). By 2023, systematic reviews documented over 300 ML-based yield-prediction studies, with Random Forest and Gradient Boosting consistently among the top-performing algorithms across crops and regions (Khaki & Wang, 2023). The intersection of these two

trajectories, USSD delivery and ML-based prediction, remains under-explored, motivating the present research.

2.3 Overview of Existing Systems

Five categories of software systems currently address parts of the problem space targeted by this research.

- e-Soko (Rwanda, MINAGRI 2012-present). A USSD- and web-based platform that publishes daily commodity prices across Rwandan markets. It is widely used but offers neither weather forecasting nor yield prediction (MINAGRI, 2021).
- PlantVillage Nuru (Penn State / CGIAR, 2018-present). An Android application that uses convolutional neural networks to diagnose crop diseases from images. Achieves >90% accuracy on cassava mosaic disease but requires a smartphone with a camera and is not localised for Rwandan beans or maize (Hughes et al., 2021).
- iCow (Kenya, Green Dreams 2010-present). A USSD and SMS service providing livestock keeping calendars. Rule-based, with no ML personalisation.
- Esoko Insyt (Ghana, Esoko 2018-present). A mobile data-collection and advisory platform combining SMS, voice and a smartphone app. Includes weather alerts but uses generic forecast bulletins.
- FAO WaPOR (FAO, 2018-present). A web dashboard providing satellite-derived water productivity indicators. Powerful for analysts but requires internet access and technical literacy; not designed for end-user farmers.

2.4 Review of Related Work

van Klompenburg, Kassahun and Catal (2020) conducted a systematic literature review of 50 ML-based yield-prediction studies between 2008 and 2019 and concluded that Random Forest and Gradient Boosting delivered the most consistent performance, with neural networks outperforming when large multispectral inputs were available. Their review explicitly noted the absence of deployment studies targeting smallholder farmers in resource-constrained settings.

Khaki, Wang and Archontoulis (2020) proposed a CNN-RNN hybrid for county-level maize

and soybean yield prediction in the United States, achieving an RMSE of 9% of mean yield. While the model architecture is powerful, its data requirements (USDA-grade remote sensing and high-resolution soil grids) exceed what is available for Rwandan districts.

Cedric et al. (2022) trained Random Forest, Decision Tree and Bayes Net classifiers on a multi-country African crop dataset and reported classification accuracies up to 99.6% for crop suitability tasks. While the figure is encouraging, the study used a discretised classification target rather than continuous yield, limiting its applicability to harvest-quantity forecasting.

Sharma, Saha and Wadhwa (2023) demonstrated that a Random Forest regressor trained on weather and soil features predicted wheat yield with R squared of 0.81 in Indian agro-climatic zones, and crucially discussed model deployment on edge devices for offline inference, a principle directly applicable to the present work.

Singh and Patel (2024) integrated meteorological data and pesticide-use logs to forecast maize yield in IEEE Access, achieving MAE within 12% of mean yield. Their feature-engineering approach, particularly the use of cumulative growing-degree-days, informs the feature design in Chapter 3 of this proposal.

On the delivery side, Baumuller (2018) evaluated mobile-phone-based agricultural extension in Kenya and reported that text-message subscribers increased their use of recommended inputs by 22%, demonstrating measurable behavioural impact from low-tech channels. Aker, Ghosh and Burrell (2016) similarly found that SMS-delivered agronomic information improved adoption of recommended practices among Nigerien farmers.

Closer to the Rwandan context, Mugisha and Diaz (2021) profiled USSD service adoption in East Africa and reported that mobile-money usage exceeds 80% among rural adults across the region, validating USSD as a viable delivery channel. Nakayima and Tumwine (2022) piloted a USSD-based fertiliser recommendation service in Uganda with 200 farmers and reported a 73% retention rate over three months, providing encouraging benchmarks for the present pilot design.

Finally, Tessema, Mengistu and Hailu (2023) presented an MLOps framework for agricultural ML services in Ethiopia that emphasises model versioning, drift monitoring and gradual rollout, considerations incorporated into the system architecture described in Chapter 3.

2.5 Summary of the Reviewed Literature

#	Author(s) & Year	Focus	Key Finding	Limitation
1	van Klompenburg et al. (2020)	Systematic review of ML yield	Random Forest and Gradient Boosting	No deployment focus on

		prediction	dominate	smallholders
2	Khaki et al. (2020)	CNN-RNN for US maize/soybean	RMSE 9% of mean yield	Requires high resolution remote sensing
3	Cedric et al. (2022)	RF, DT, Bayes Net for African crops	Up to 99.6% suitability accuracy	Classification not regression
4	Sharma et al. (2023)	RF for Indian wheat yield	R squared = 0.81; edge deployment	Different agro climatic zone
5	Singh & Patel (2024)	Weather + pesticide for maize	MAE within 12% of mean yield	Pesticide data unavailable in Rwanda
6	Baumuller (2018)	SMS extension in	+22% input adoption	No predictive component

		Kenya		
7	Aker et al. (2016)	SMS agronomic info in Niger	Behavioural change documented	Static, rule-based content
8	Hughes et al. (2021)	PlantVillage Nuru CNN	>90% disease detection accuracy	Smartphone-onl y
9	Mugisha & Diaz (2021)	USSD adoption in East Africa	80% rural mobile money usage	Descriptive, not prescriptive
10	Nakayima & Tumwine (2022)	USSD fertiliser service in Uganda	73% three-month retention	Rule-based, no ML
11	Khaki & Wang (2023)	Yield-predic tion review update	Ensembles still leading	Limited LMIC case studies
12	Tessema et al. (2023)	MLOps for ag-ML in Ethiopia	Drift monitoring essential	No farmer-facing UI evaluation

Table 2.1: Summary of Reviewed Literature.

2.6 Strengths and Weaknesses of the Existing Systems

2.6.1 Strengths of Current Systems

- The wide reach of SMS- and USSD-based platforms (e-Soko, iCow, Esoko) demonstrates that GSM-only channels can sustain national-scale agricultural information services.
- ML-based prediction models (Random Forest, XGBoost, CNN-RNN) achieve documented accuracies of 80-95% in controlled settings, validating the technical feasibility of yield forecasting.
- Smartphone applications such as PlantVillage Nuru showcase the potential of computer vision and CNNs for disease diagnosis.
- MLOps and edge-deployment patterns are mature enough to support low-latency inference in production.

2.6.2 Weaknesses and Limitations

- Existing USSD/SMS platforms are predominantly rule-based and do not integrate predictive ML models.
- Smartphone-only solutions systematically exclude feature-phone users, who represent the majority of Rwandan smallholders.
- Many published ML studies report accuracy on retrospective splits but lack a deployment artefact or usability evaluation.
- Few solutions support local-language interfaces beyond English and French.
- No reviewed system simultaneously combines: (a) feature-phone accessibility, (b) ML-based yield prediction, (c) localised language support, and (d) integration with a national meteorological service.

System	Strength	Weakness
e-Soko	Wide adoption, free for farmers	Prices only; no weather or yield
iCow	Bilingual, USSD-based	Livestock focus; rule-based
Esoko Insyt	Multi-channel (SMS/voice/app)	Generic forecasts; no per-plot model

PlantVillage Nuru	High-accuracy disease CNN	Requires smartphone and camera
FAO WaPOR	Authoritative satellite data	Analyst-oriented dashboard

Table 2.2: Strengths and Weaknesses of Existing Systems.

2.6.3 Comparison of this System with Current Tools

Capability e-Soko iCowProposed System
PlantVillage Nuru

Feature-phone access (USSD)	Yes	Yes	No	Yes
Local-language UI (Kinyarwanda)	Partial	No	No	Yes
Personalised yield prediction (ML)	No	No	No	Yes
Localised weather integration	No	Partial	No	Yes
Crop disease diagnostics	No	No	Yes	No (future work)
Works without mobile data	Yes	Yes	No	Yes
Open / extensible architecture	No	No	Partial	Yes

Table 2.3: Comparative Analysis: Proposed System vs Existing Tools.

2.7 General Comment and Conclusion

The reviewed literature converges on three robust findings: ensemble machine-learning regressors are mature enough to deliver useful yield predictions for staple crops; low-bandwidth channels such as USSD and SMS achieve broader rural reach than smartphone applications; and behavioural impact studies confirm that timely information meaningfully changes farmer decisions. However, no single system integrates these three findings into a deployed, evaluated artefact for Rwandan smallholders. The proposed system targets precisely this gap, contributing both a software engineering deliverable and an empirical evaluation of the USSD + ML pairing in an East African context.

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter describes the methodological and architectural choices that translate the objectives of Chapter 1 and the gaps identified in Chapter 2 into an implementable software system. The design approach combines (a) an iterative agile-scrum SDLC adapted to the nine week project window, (b) a multi-tier modular system architecture, and (c) UML modelling to specify behaviour and data.

3.2 Research Design

The research adopts a design-science methodology, treating the USSD-based advisory system as the central artefact whose construction and evaluation generate the knowledge contribution (Hevner et al., 2004). Both quantitative and qualitative data are collected: quantitative model accuracy metrics (R squared, RMSE, MAE) from retrospective validation, and qualitative usability data from structured interviews and SUS questionnaires administered during the pilot.

3.2.1 Model of Development Framework

An iterative agile-scrum SDLC is adopted in two-week sprints. Each sprint completes a Plan Design-Build-Test-Review cycle, with the supervisor acting as product owner. The model favours rapid feedback incorporation, which is critical given the compressed timeline and the user-research nature of the pilot phase.

Figure 8. Iterative Agile-Scrum Development Model Adapted for the Project

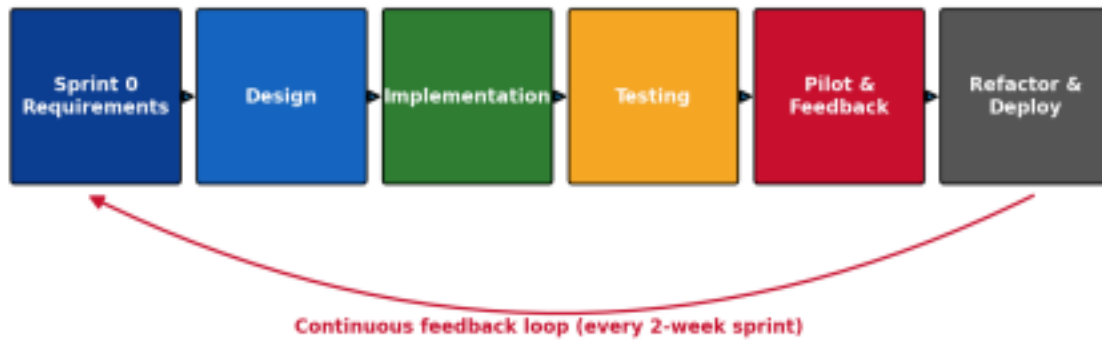


Figure 8: Iterative Agile-Scrum Development Model Adapted for the Project.

3.3 Requirements Specification

3.3.1 Functional Requirements

ID	Requirement	Priorit y
FR-01	The system shall expose a USSD shortcode through which a farmer can initiate a session from any GSM handset.	Must
FR-02	The system shall authenticate farmers by their MSISDN and store basic profile data (district, language, crop).	Must
FR-03	The system shall display a top-level menu in either English or Kinyarwanda, switchable at any time.	Must
FR-04	The system shall fetch a seven-day weather forecast for the farmer's district from the Rwanda Meteorology Agency API.	Must

FR-05	The system shall accept a planting date, plot size and crop selection and return an ML-predicted yield with a confidence band.	Must
FR-06	The system shall provide planting-window advice derived from forecasted rainfall and growing-degree-day estimates.	Should
FR-07	The system shall log every USSD session for analytics and model retraining.	Should
FR-08	The system shall allow extension officers to review aggregated farmer logs via a web dashboard.	Should
FR-09	The system shall send SMS fallback alerts when severe weather is forecast.	Could
FR-10	The system shall capture pest-symptom selections and return rule-based advisory text.	Could

Table 3.1: Functional Requirements (MoSCoW prioritisation).

3.3.2 Non-Functional Requirements

ID	Category	Requirement
NFR 01	Performance	USSD round-trip latency shall not exceed 2 seconds at the 95th percentile.
NFR 02	Availability	Backend availability shall meet or exceed 99% during pilot hours (06:00-20:00 CAT).

NFR 03	Usability	The USSD menu shall require no more than four menu levels for any advisory task.
NFR 04	Localisation	All user-facing text shall be available in English and Kinyarwanda.
NFR 05	Security	Admin endpoints shall require JWT authentication; passwords shall be hashed with bcrypt.
NFR 06	Privacy	MSISDN shall be hashed in analytics exports; raw numbers stored only with farmer consent.
NFR 07	Maintainability	Code shall achieve at least 70% line coverage in automated tests.
NFR 08	Portability	The system shall be deployable via Docker on any cloud provider.

Table 3.2: Non-Functional Requirements.

3.4 System Architecture

Figure 1 depicts the four-tier architecture: presentation, application, data-and-model, and external data sources. The presentation tier exposes the system to farmers through telecom USSD gateways and, optionally, to extension officers through a web dashboard. The application tier hosts a FastAPI USSD session handler and a hybrid rules-plus-ML advisory engine. The data and-model tier persists operational data in PostgreSQL and serves trained ML artefacts from a model registry. The external tier integrates the Rwanda Meteorology Agency API, historical datasets from NISR and MINAGRI, FAO agro-climatic references and OpenWeather as a backup

forecast source.

Figure 1. System Architecture of the USSD-Based Agricultural Advisory System

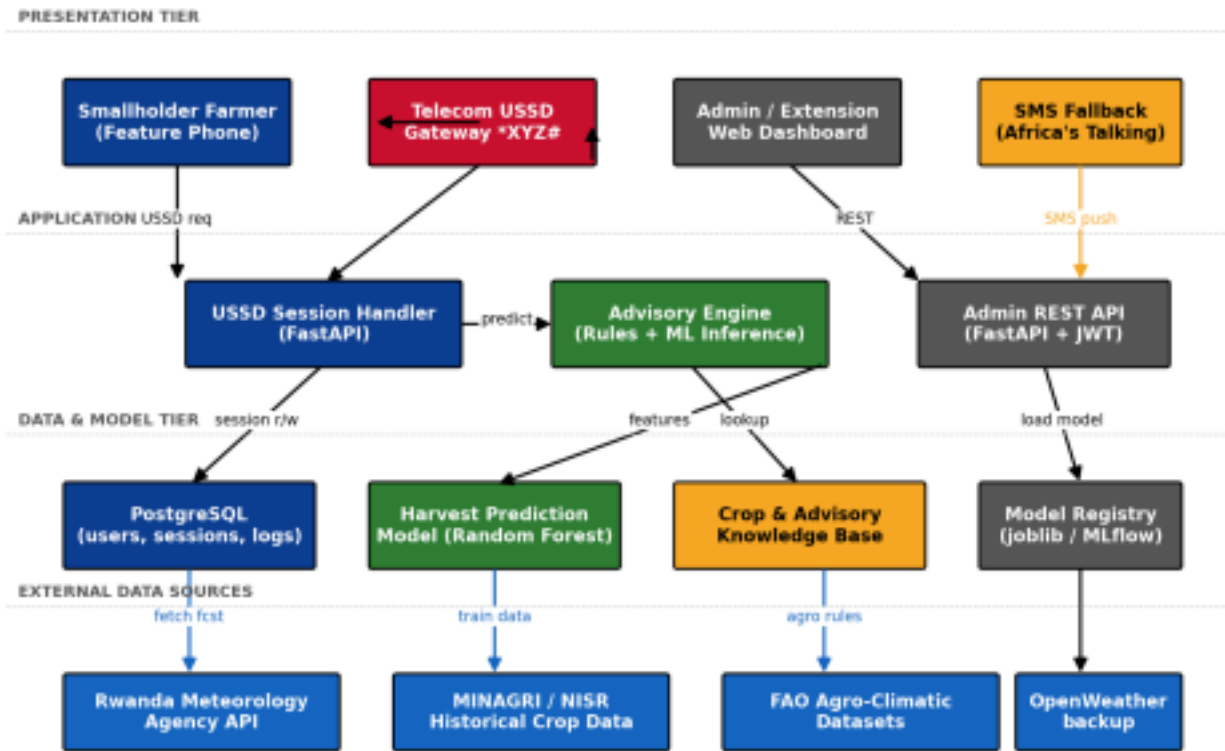


Figure 1: System Architecture of the USSD-Based Agricultural Advisory System.

3.4.1 Data Definition and Acquisition

The project adopts a multi-source data strategy combining three categories of inputs.

- Historical yield data: bean and maize yields per district per season from 2015-2024, sourced from NISR Statistical Yearbooks and MINAGRI's Seasonal Agricultural Survey (estimated 4,500-6,000 rows after cleaning).
- Weather data: daily rainfall, minimum and maximum temperature, humidity and evapotranspiration from the Rwanda Meteorology Agency for Nyagatare and adjacent districts (2015-2024).
- Crop and agro-climatic references: FAO crop water-requirement tables, growing-degree-day coefficients and disease-trigger thresholds for the targeted crops.

Each dataset is documented with provenance, licence and any privacy constraints; raw files are stored in object storage with checksums to ensure reproducibility.

3.4.2 Machine Learning Training Pipeline

The pipeline is implemented in Python using scikit-learn and XGBoost. Its stages are: (i) ingestion of raw data; (ii) cleaning, deduplication and timezone normalisation; (iii) feature engineering, including cumulative growing-degree-days, rainfall-week aggregates, and historical district-level yield averages; (iv) train-test split with stratification by season; (v) cross-validated training of candidate models with hyper-parameter tuning via Optuna; (vi) evaluation against held-out 2024 yields; and (vii) export of the selected model artefact to the registry. The expected output is a joblib-serialised regression model returning predicted yield (kg per are) with a 90% confidence band.

3.4.3 Model Comparison and Architecture

Three supervised regression families are benchmarked. Defaults are explicitly avoided; the final selection is justified by hold-out performance, inference latency and explainability.

Algorithm	Rationale	Library	Expected R squared
Random Forest Regressor	Strong baseline on tabular agronomic data; robust to missing values	scikit-learn	0.65 - 0.75
XGBoost Regressor	Documented top performance on similar yield problems; supports SHAP explanations	xgboost	0.70 - 0.82
Multilayer Perceptron	Tests whether non-linear interactions captured by NN improve over trees	scikit-learn	0.60 - 0.78

Table 3.3: Candidate ML Algorithms for Yield Prediction.

Model selection is finalised on the basis of R squared and MAE on the 2024 hold-out, with inference time and interpretability as secondary criteria.

3.5 UML Diagrams

3.5.1 Use Case Diagram

Figure 3 identifies four actors (Farmer, Extension Officer, Admin, Meteorology Agency API) and eleven primary use cases, ranging from the farmer-facing 'Get harvest prediction' to administrative functions such as 'Retrain ML model' and 'Manage users and roles'.

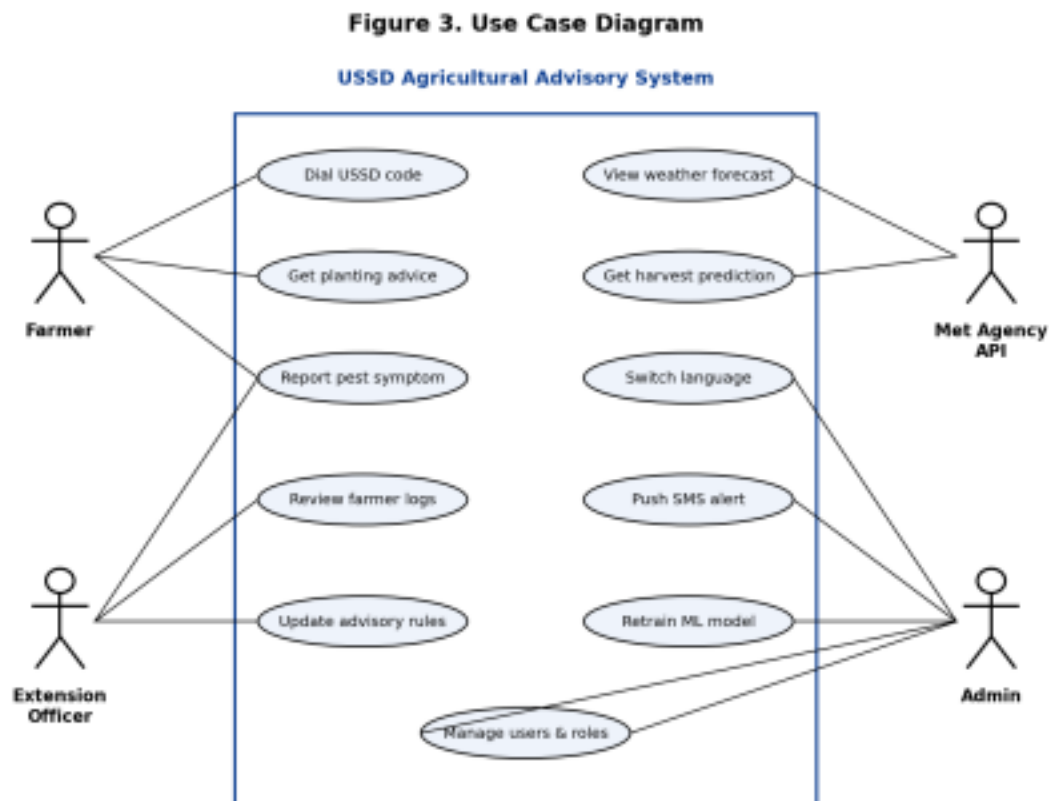


Figure 3: Use Case Diagram.

3.5.2 Class Diagram

Figure 4 specifies the principal domain classes and their relationships. The Farmer aggregates many USSDSession instances; each session contains one or more AdvisoryRequest objects; each request resolves to exactly one HarvestForecast generated by a PredictionModel. WeatherSnapshot, Crop and AdvisoryRule support the advisory engine, while AdminUser and SMSAlert handle operations.

Figure 4. Class Diagram (UML)

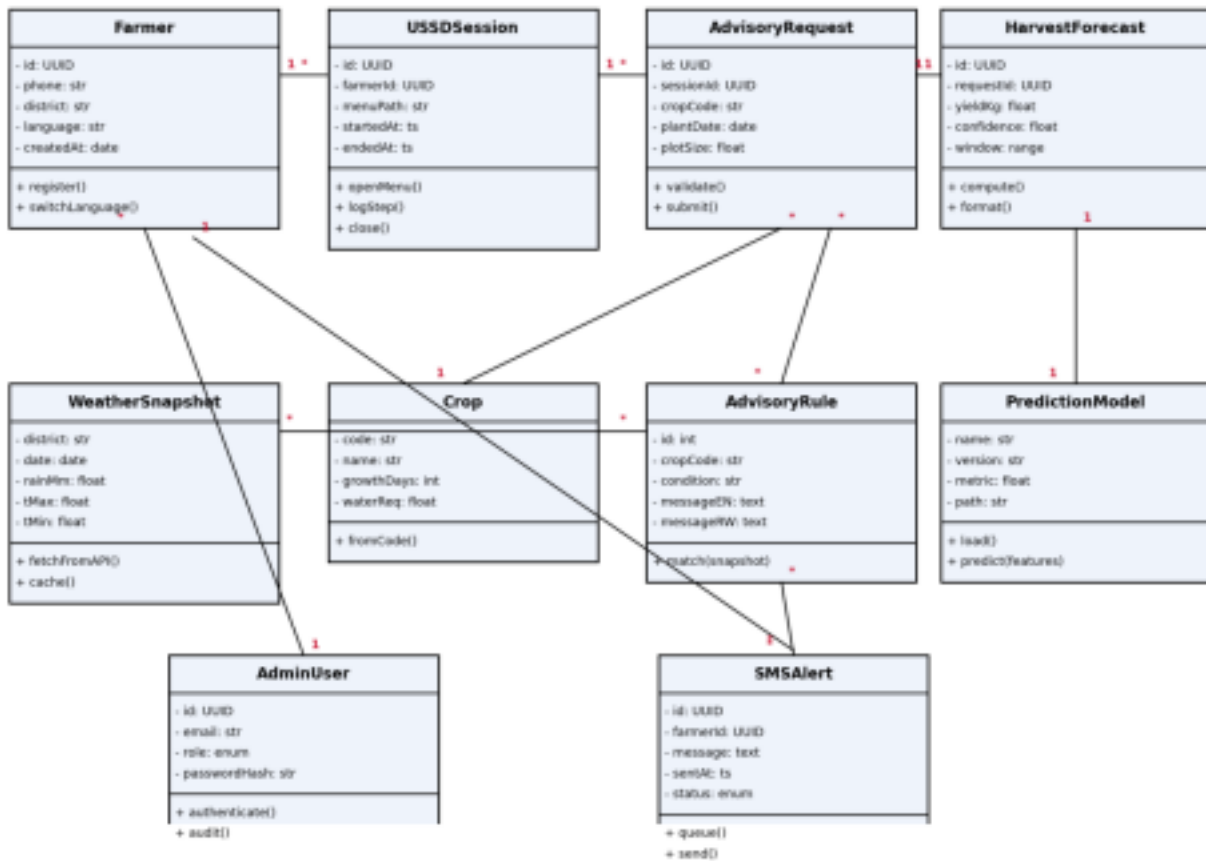


Figure 4: Class Diagram (UML).

3.5.3 Entity-Relationship Diagram

Figure 5 shows the relational schema in Crow's-foot notation. The schema normalises farmer, district, session, request, forecast, weather snapshot, crop, advisory rule and SMS alert entities, with primary and foreign keys identified.

Figure 5. Entity-Relationship Diagram (Crow's-Foot Notation)

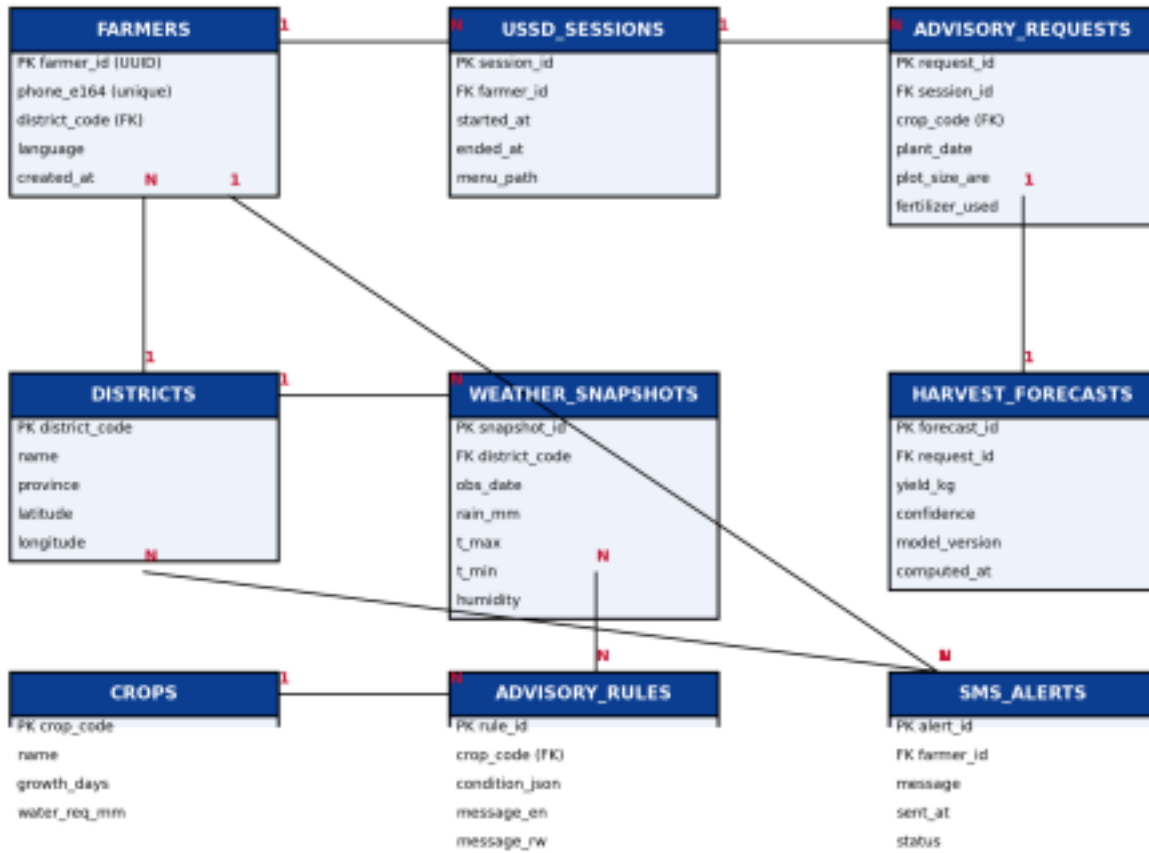


Figure 5: Entity-Relationship Diagram (Crow's-Foot Notation).

3.5.4 Sequence Diagram

Figure 6 traces the most performance-critical interaction: a farmer's harvest-forecast request, from the initial USSD dial through the telecom gateway, USSD handler, advisory engine, weather cache, prediction model and database, ending with the rendered text message on the handset.

Figure 6. Sequence Diagram: Harvest Forecast Request

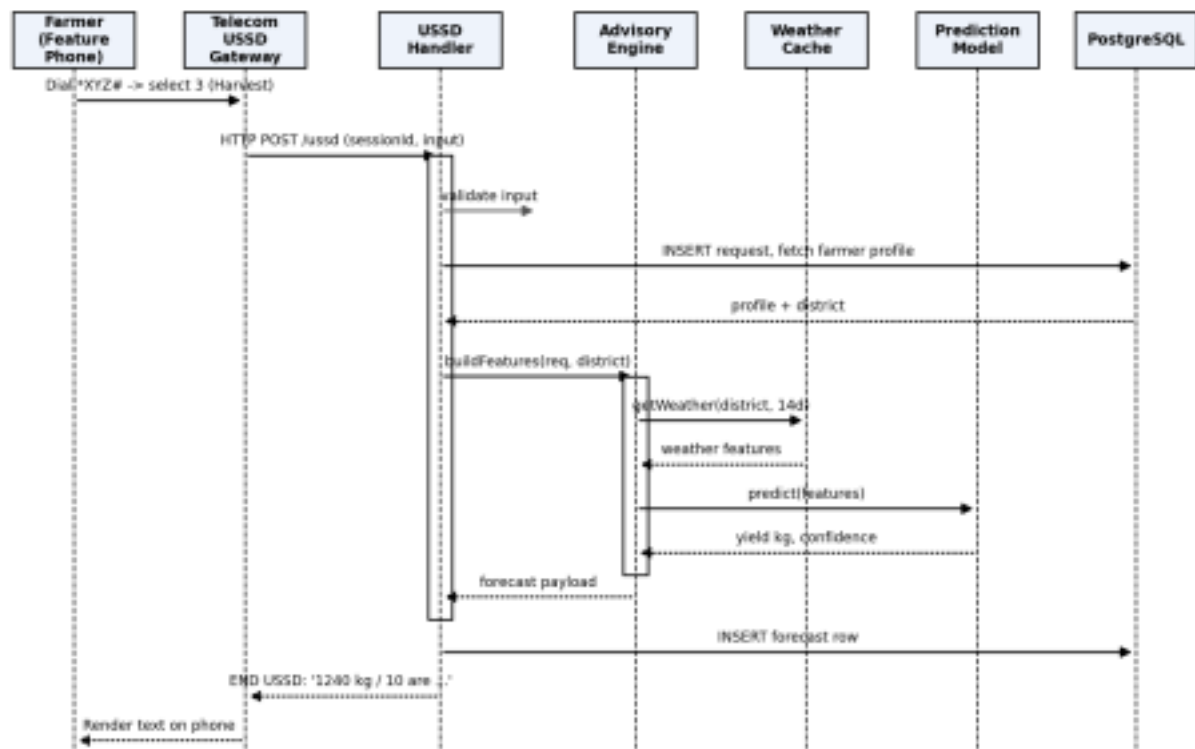


Figure 6: Sequence Diagram: Harvest Forecast Request.

3.5.5 USSD Menu Flow

Figure 2 details the proposed farmer-facing menu hierarchy. The top-level menu offers weather, planting-date advice, harvest forecast, pest help, profile management and language switching. Each branch is constrained to a maximum of four levels in line with NFR-03.

Figure 2. USSD Menu Flow for the Farmer Advisory Service

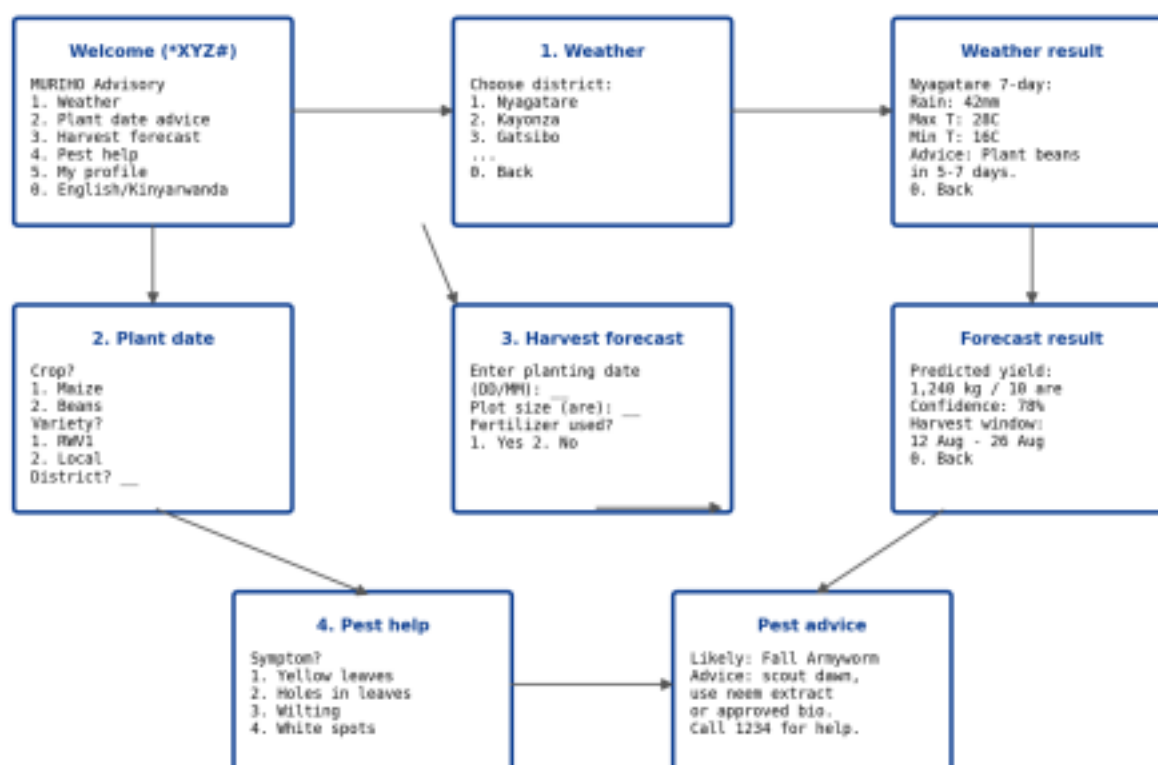


Figure 2: USSD Menu Flow for the Farmer Advisory Service.

3.6 Development Tools

Layer	Tool / Technology	Justification
USSD gateway	Africa's Talking sandbox	Free tier; widely used in East Africa; REST callback model
Backend language / framework	Python 3.11 + FastAPI	Asynchronous, low-latency; aligns with ML stack
ML libraries	scikit-learn, XGBoost, Optuna	Industry standard; supports the regressors in Table 3.3

Database	PostgreSQL 16	Mature, open source; supports JSONB for advisory rules
Caching	Redis (free tier)	Weather-snapshot caching to meet 2 s latency
Containerisation	Docker + Docker Compose	Portable deployment across providers
Hosting	Render / Railway (free + paid)	Free tier sufficient for pilot

Dashboard	Streamlit	Rapid build for the extension-officer view
Version control	Git + GitHub	Industry standard; supports CI
CI / CD	GitHub Actions	Free for student / open-source repositories
Diagramming	Matplotlib / draw.io	Reproducible diagrams under version control
Project tracking	GitHub Projects + Notion	Agile boards aligned with sprint cadence

Table 3.4: Development Tools and Justification.

REFERENCES

- Aker, J. C., & Fafchamps, M. (2015). Mobile phone coverage and producer markets: Evidence from West Africa. *World Bank Economic Review*, 29(2), 262-292.
- Aker, J. C., Ghosh, I., & Burrell, J. (2016). The promise (and pitfalls) of ICT for agriculture initiatives. *Agricultural Economics*, 47(S1), 35-48.
- Aker, J. C., & Mbiti, I. M. (2010). Mobile phones and economic development in Africa. *Journal of Economic Perspectives*, 24(3), 207-232.
- Baumuller, H. (2018). The little we know: An exploratory literature review on the utility of mobile phone-enabled services for smallholder farmers. *Journal of International Development*, 30(1), 134-154.
- Cedric, L. S., Adoni, W. Y. H., Aworka, R., Zoueu, J. T., Mutombo, F. K., Krichen, M., & Kimpolo, C. L. M. (2022). Crops yield prediction based on machine learning models: Case of West African countries. *Smart Agricultural Technology*, 2, 100049.
- Food and Agriculture Organization of the United Nations. (2021). The state of food and agriculture 2021: Making agrifood systems more resilient to shocks and stresses. FAO.
<https://www.fao.org/publications/sofa/2021/en/>
- Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. *MIS Quarterly*, 28(1), 75-105.
- Hughes, D. P., Salathe, M., et al. (2021). AI for agriculture: PlantVillage Nuru. *Frontiers in Plant Science*, 12, 647383. <https://doi.org/10.3389/fpls.2021.647383>
- Intergovernmental Panel on Climate Change. (2022). Climate change 2022: Impacts, adaptation and vulnerability. Cambridge University Press. <https://www.ipcc.ch/report/ar6/wg2/>
- Khaki, S., & Wang, L. (2023). Machine learning for crop yield prediction: A review of methods and applications. *Computers and Electronics in Agriculture*, 207, 107755.
- Khaki, S., Wang, L., & Archontoulis, S. V. (2020). A CNN-RNN framework for crop

yield prediction. *Frontiers in Plant Science*, 10, 1750.

Lipper, L., Thornton, P., Campbell, B. M., Baedeker, T., Braimoh, A., Bwalya, M., ... & Torquebiau, E. F. (2014). Climate-smart agriculture for food security. *Nature Climate Change*, 4(12), 1068-1072.

Ministry of Agriculture and Animal Resources [MINAGRI]. (2021). Agricultural digital initiatives report 2020-21. Government of Rwanda.

Mugisha, J., & Diaz, R. (2021). USSD service adoption among rural populations in East Africa: A multi-country analysis. *African Journal of Information Systems*, 13(2), 87-105.

Nakayima, A., & Tumwine, S. (2022). A USSD-based fertiliser recommendation service for smallholder farmers in Uganda. *International Journal of Agricultural Informatics*, 14(3), 211- 228.

National Institute of Statistics of Rwanda [NISR]. (2023). Statistical yearbook 2023. Government of Rwanda. <https://www.statistics.gov.rw/>

Rwanda Meteorology Agency. (2022). Downscaled climate projections for Rwanda. Government of Rwanda. <https://www.meteorwanda.gov.rw/>

Rwanda Utilities Regulatory Authority [RURA]. (2023). Telecommunications statistics report Q4 2023. Government of Rwanda. <https://www.rura.rw/>

Sharma, A., Saha, S., & Wadhwa, P. (2023). Random forest based yield prediction with edge deployment for Indian wheat-growing regions. *Precision Agriculture*, 24(4), 1320-1342.

Singh, R., & Patel, M. (2024). Incorporating meteorological data and pesticide information to forecast crop yields using machine learning. *IEEE Access*, 12, 11874-11890.

Tessema, T., Mengistu, A., & Hailu, B. (2023). An MLOps framework for agricultural machine learning services in Ethiopia. *Journal of African Earth Sciences*, 198, 104810.

van Klompenburg, T., Kassahun, A., & Catal, C. (2020). Crop yield prediction using machine learning: A systematic literature review. *Computers and Electronics in*

Agriculture, 177, 105709.